A

REAL TIME RESEARCH

PROJECT REPORT

ON

University Student Academic Performance & Engagement Analytics Dashboard

SIDDHARTHA INSTITUTE OF TECHNOLOGY & SCIENCES
(UGC – AUTONOMOUS)

(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)

Accredited by NBA and NAAC with 'A+' Grade.

Narapally, Korremula Road, Ghatkesar, Medchal- Malkajgiri (Dist.)-500088

(Submitted in partial fulfilment of the academic requirements of B. Tech) In

DEPARTMENT OF CSE (DATA SCIENCE)

By

S.GANESH (24TQ1A6742)

R.ARUN (24TQ1A6762)

K.MAHESH (24TQ1A6727)

S.RAMCHARAN (24TQ1A6740)

Under the Esteemed Guidance of

Ms.J.Priyanka

SIDDHARTHA INSTITUTE OF TECHNOLOGY AND SCIENCES
(UGC – AUTONOMOUS)

(Approved by AICTE, Affiliated to JNTU Hyderabad, Accredited by NAAC(A+))

Korremula Road, Narapally(V), Ghatkesar Mandal, Medchal-Dist:-500088

CERTIFICATE

This is to certify that the project report entitled

University Student Academic Performance & Engagement Analytics Dashboard

being submitted

by

S. GANESH	**(24TQ1A6742)**
R.ARUN	**(24TQ1A6762)**
K.MAHESH	**(24TQ1A6727)**
S.RAMCHARAN	**(24TQ1A6740)**

In partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science), Jawaharlal Nehru Technological University Hyderabad, is a record of bonafide work carried out under my guidance and supervision. The results embodied in this Realtime Research project report have not been submitted to any other University or Institute for the award of any Degree or Diploma

Guide	Co-Ordinator	Head of the department
Ms.J.Priyanka	**Mr.K.VIJAY**	**Mr.K.VIJAY**
Department of CSE(DS)	(Assistant Professor)	Department of CSE(DS)

Internal Examiner External Examiner

DECLARATION

We declare that this project report titled:

University Student Academic Performance & Engagement Analytics Dashboard subm itted in partial fulfilment of the degree of **B. Tech in CSE(DATA SCIENCE)** is a record of original work carried out by us under the supervision of **Ms.J.Priyanka** and has not formed the basis for the award of any other degree or diploma, in this or any other Institute or University. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

By

S. GANESH	**(24TQ1A6742)**
R.ARUN	**(24TQ1A6762)**
K.MAHESH	**(24TQ1A6727)**
S.RAMCHARAN	**(24TQ1A6740)**

ACKNOWLEDGEMENT

Any endeavor in the field of development is a person's intensive activity. A successful project is a fruitful culmination of efforts by many people, some directly involved and some others who have quietly encouraged and supported.

Salutation to be beloved and highly esteemed institute SIDDHARTHA INSTITUTE OF TECHNOLOGY & SCIENCES for grooming us into Computer Science and Engineering(Data Science) graduate, We wish to thank PRINCIPAL **DR. M. JANARDHAN** ,

VICE PRINCIPAL **DR.G.PRAVEEN KUMAR** for providing a great learning environment.

We wish to express profound gratitude to **MR.K.VIJAY**, Assistant Professor and HEAD OF DEPARTMENT**,** CSE(DATA SCIENCE), for her continuous encouragement to ensure successful results in all my endeavors.

We would like to thank **Ms.J.Priyanka**, Department of CSE (DATA SCIENCE), who patiently guided and helped us throughout our Realtime Research project.

We take this opportunity to thank the department's Project Review Co- Ordinator **MR.K.VIJAY** for all the review meetings, suggestions, and support throughout the project development. .

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Vision of the Department: To be a Recognized Center of Computer Science Education with values and quality research.

Mission of the Department:

MISSION	STATEMENT
DM1	Import High Quality Professional Training With An Emphasis On Basic principles Of Computer Science And Allied Engineering
DM2	Imbibe Social Awareness And Responsibility To Serve The Society.
DM3	Provide Academic Facilitates Organize Collaborated Activities To enable Overall Development Of Stakeholders

Programme Educational Objectives (PEO)

- **PEO1:** Graduates will be able to synthesize mathematics, science, engineering fundamentals, laboratory and work – based experiences to formulate and to solve problems proficiently in Computer science and Engineering and related domains.
- **PEO2:** Graduates will be prepared to communicate effectively and work in multidisciplinary engineering projects following the ethics in their profession.
- **PEO3:** Graduates will recognize the importance of and acquire the skill of independent learning to shine as experts in the field with a sound knowledge.

ABSTRACT

This project presents the development of a *University Student Academic Performance and Engagement Analytics Dashboard* using Power BI, aimed at analyzing and visualizing key factors influencing student success. The dashboard consolidates data related to academic performance, attendance, and student engagement to provide a comprehensive view of student behavior and outcomes.

1. By leveraging Power BI's data modeling and interactive visualization capabilities, the system transforms raw data into meaningful insights through dynamic reports and dashboards. It enables users to track key performance indicators, identify trends, and detect students who may be at risk of poor academic performance. Features such as filters, drill-down analysis, and real-time updates enhance usability and support informed decision-making.

TABLE OF CONTENTS

DESCRIPTION	PAGE NUMBER
CERTIFICATE	**II**
DECLARATION	**III**
ACKNOWLEDGEMENT	**IV**
ABSTRACT	**VI**
CHAPTER 1: INTRODUCTION	10
1.1 PROBLEM DEFINATION	11
1.2 OBJECTIVE OF PROJECT	11
1.3 SCOPE OF THE PROJECT	11
1.4 MOTIVATION	12
CHAPTER 2: LITERATURE SURVEY	13
CHAPTER 3: EXISITING SYSTEM	18
3.1 DRAWBACKS OF EXISTING SYSTEM	20
CHAPTER 4: PROPOSED SYSTEM	21
CHAPTER 5: HARDWARE AND SOFTWARE REQUIREMENTS	
5.1 HARDWARE REQUIREMENTS	23
5.2 SOFTWARE REQUIREMENTS	24
CHAPTER 6: SYSTEM DESIGN	
6.1 SYSTEM DESIGN	25
6.2 UML DIAGRAMS	27
CHAPTER 7: IMPLEMENTATION	**30**
7.1 SOLIDITY SOURCE CODE	33

7.2 JAVASCRIPT SOURCE CODE 34

7.3 HTML SOURCE CODE 37

CHAPTER 8: RESULTS 38

CHAPTER 9: CONCLUSION 42

CHAPTER 10: FUTURE SCOPE 44

CHAPTER 11: REFERENCES 46

CHAPTER 1
INTRODUCTION

1. INTRODUCTION

In recent years, the use of data analytics in the education sector has gained significant importance, as institutions strive to improve student outcomes and overall academic performance. Universities generate large volumes of data related to student attendance, grades, participation, and engagement. However, without proper tools, this data often remains underutilized and fails to support effective decision-making.

This project, *University Student Academic Performance and Engagement Analytics Dashboard using Power BI*, aims to address this challenge by providing a comprehensive and interactive platform for analyzing

student data. By leveraging Power BI, a powerful business intelligence tool, the project transforms raw academic and engagement data into visually intuitive dashboards and reports.

The dashboard enables educators and administrators to monitor key performance indicators such as student grades, attendance patterns, and engagement levels. It also helps in identifying trends, comparing performance across different groups, and detecting at-risk students who may require additional support. The use of interactive features such as filters and drill-down capabilities enhances the user experience and allows for deeper data exploration.

Overall, this project demonstrates how data visualization and analytics can play a vital role in improving educational strategies, enhancing student engagement, and supporting data-driven decision-making in universities.

1.1 PROBLEM DEFINITION

In many universities, large amounts of student data—such as academic performance, attendance, and engagement—are collected regularly. However, this data is often stored in separate systems or formats, making it difficult to analyze effectively. As a result, educators and administrators face challenges in gaining clear and unified understanding of student performance.

1.2 OBJECTIVE OF PROJECT

The main objective of this project is to develop an interactive *University Student Academic Performance and Engagement Analytics Dashboard* using Power BI to analyze and visualize student data effectively

1.3 SCOPE OF PROJECT

The scope of this project focuses on the design and development of a *University Student Academic Performance and Engagement Analytics Dashboard* using Power BI to provide meaningful insights into student data.

This project includes the collection, integration, and analysis of student-related data such as academic scores,attendance records, and engagement metrics. The dashboard is designed to present this data through interactive visualizations, enabling users to easily monitor and evaluate student performance

The project is limited to data visualization and analysis using Power BI and does not include predictive modeling or automated intervention systems. It relies on the availability and quality of the input data provided.

The system allows educators and administrators to:

- Track key performance indicators (KPIs) related to academic success.
- Analyze trends and patterns in student performance and engagement.
- Identify students who may be at risk and require additional support.

1.3 MOTIVATION

The motivation behind this project arises from the growing need to effectively utilize student data to improve academic performance and engagement in universities. Although large volumes of data are generated through academic activities, attendance systems, and student interactions, much of this data is not fully analyzed or used for decision-making.

Educators often rely on traditional and manual methods to evaluate student performance, which can be time-consuming and may not provide accurate or timely insights. This makes it difficult to identify struggling students early and implement necessary interventions. Additionally, the lack of clear and interactive visualizations limits the ability to understand complex data patterns.

With the advancement of business intelligence tools like Power BI, there is an opportunity to transform raw data into meaningful and interactive visual insights. This project is motivated by the desire to create a user-friendly dashboard that simplifies data analysis, enhances visibility into student performance, and supports data-driven decisions.

Ultimately, the goal is to help educational institutions improve teaching strategies, increase student engagement, and ensure better academic outcomes through the effective use of analytics.

CHAPTER 2
LITERATURE SURVEY

1. LITERATURE SURVEY

The use of data analytics and business intelligence in education has gained significant attention in recent years. Researchers and institutions have explored various methods to analyze student performance and engagement in order to improve academic outcomes.

Several studies highlight the importance of analyzing academic data such as grades, attendance, and participation to identify patterns that influence student success. Traditional statistical methods have been widely used; however, they often lack interactivity and real-time insights. With the advancement of data analytics tools, there has been a shift toward more dynamic and visual approaches

Business Intelligence (BI) tools, particularly Power BI, have been increasing adopted in the education sector due to their ability to integrate data from multiple sources and present it through interactive dashboards. These dashboards help educators monitor key performance indicators (KPIs), track student progress, and make informed decisions. Research shows that visual analytics improves the understanding of complex data and enhances decision-making efficiency.

Some studies also emphasize the role of early identification of at-risk students using data analytics. By analyzing engagement metrics and academic performance, institutions can take proactive measures to support students and reduce dropoutrates. However, many existing systems are either complex, costly, or lack user-friendly interfaces, making them less accessible for everyday use in educational institutions.

This project addresses these gaps by developing a simple, interactive, and efficient analytics dashboard using Power BI.

In conclusion, the literature indicates that integrating data analytics and visualization tools in education can significantly improve student performance, engagement, and institutional decision-making.

This project builds upon these research findings by developing a Power BI-based Analytics dashboard that is simple, interactive, and efficient. It focuses on integrating academic performance and engagement data into a single platform, enabling better visualization, analysis and decision-making.

In conclusion, the literature strongly supports the use of data analytics and visualization tools in education, highlighting their potential to improve student outcomes, enhance engagement, and support institutional planning.

CHAPTER 3
EXISTING SYSTEM

2. EXISTING SYSTEM

1 . Library-based implementations

Library-based implementations refer to traditional and pre-built systems used in educational institutions to manage and analyze student data. These systems often rely on predefined libraries, tools, or software modules that provide basic functionalities such as data storage, retrieval, and simple reporting.

2. Educational tutorials and notebooks

Educational tutorials and notebooks are widely used as learning resources for understanding data analysis and visualization concepts. These include online tutorials, sample datasets, and interactive notebooks (such as Jupyter Notebooks) that demonstrate how to analyze student performance data using programming languages like Python or tools like Excel.

3. Academic and research examples

Academic and research examples refer to studies and prototype systems developed by researchers to analyze student academic performance and engagement using various data analytics techniques. These implementations often use methods such as statistical analysis, machine learning, and data mining to identify patterns and predict student outcomes.

4. Limitations of existing systems

Despite the availability of various methods and tools for analyzing student academic performance and engagement, existing systems have several limitations that reduce their effectiveness in real-world educational environments.

5. CONCLUSION

The *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* successfully demonstrates the importance of data analytics in the education sector. By integrating and analyzing student data related to academic performance, attendance, and engagement, the project provides valuable insights that support informed decision-making.

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The *University Student Academic Performance and Engagement Analytics Dashboard using Power BI*

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3.1 DRAWBACKS OF EXISTING SYSTEM

The existing systems used for analyzing student academic performance and engagement have several drawbacks that limit their efficiency and effectiveness in educational institutions.

One of the major drawbacks is the lack of integration among different data sources. Student information such as attendance, grades, and engagement is often stored in separate systems, making it difficult to obtain a comprehensive view of overall performance.

Another key issue is the reliance on manual data processing. Traditional methods require significant time and effort to collect, clean, and analyze data, which increases the chances of human error and delays in generating insights.

Existing systems also lack advanced visualization capabilities. Data is usually presented in static formats like tables or basic charts, which are not sufficient for understanding complex patterns and trends. This makes it harder for educators and administrators to interpret the data effectively.

Additionally, many systems are not user-friendly and require technical expertise to operate. This creates a barrier for non-technical users, limiting the accessibility and usability of the system.

The absence of real-time data analysis is another drawback. Most systems do not provide up-to-date information, making it difficult to monitor student performance continuously and take timely actions.

CHAPTER 4
PROPOSED SYSTEM

- **PROPOSED SYSTEM**

The proposed system aims to develop an interactive *University Student Academic Performance and Engagement Analytics Dashboard* using Power BI to overcome the limitations of existing systems. This system provides an efficient and user-friendly platform for analyzing and visualizing student data in a meaningful way.

- The proposed solution integrates data from multiple sources such as academic records, attendance, and engagement metrics into a single unified platform. Using Power BI, the data is cleaned, transformed, and modeled to create a structured dataset suitable for analysis.

- The dashboard presents data through interactive visualizations such as charts, graphs, and key performance indicators (KPIs). Users can easily monitor student performance, track attendance trends, and analyze engagement levels. Features

like filtering, slicing, and drill-down enable users to explore data at different levels of detail

Features

The proposed *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* offers several key features that enhance data analysis and decision-making:

- **Interactive Dashboard**
  Provides dynamic and visually appealing dashboards that allow users to interact with data easily.

- **Data Integration**
  Combines data from multiple sources such as academic records, attendance systems, and engagement metrics into a single platform.

- **Real-Time Insights**
  Enables real-time or near real-time data updates, helping users monitor student performance continuously.

- **Advanced Data Visualization**
  Displays data using charts, graphs, KPIs, and tables for better understanding of trends and patterns.

- **Filtering and Drill-Down Capabilities**
  Allows users to filter data based on specific criteria and drill down into detailed information for deeper analysis.

- **Performance Tracking**
  Tracks key performance indicators such as grades, attendance percentage, and participation levels.

Identification of At-Risk Students
Helps in detecting students who may be underperforming or less engaged, enabling early intervention.

- **User-Friendly Interface**
  Designed to be simple and easy to use, even for non-technical users like educators and administrators.

- **Scalability and Flexibility**
  Can handle large datasets and adapt to different institutional requirements.

- **Data-Driven Decision Support**
  Assists management in making informed decisions based on accurate and visualized data insights.

Benefits

The *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* provides several important benefits for educational institutions:

Better Understanding of Student Performance

Helps in clearly analyzing student marks, attendance, and engagement levels.

Data-Driven Decision Making

Enables educators and administrators to make informed decisions based on accurate data insights

- **Time Saving**
  Automates data analysis and reporting, reducing manual effort and saving time.

- **Improved Student Outcomes**
  Identifies weak or at-risk students early, allowing timely support and intervention.

- **Interactive Visualization**
  Presents data in the form of charts and dashboards, making it easy to understand complex information.

- **Real-Time Monitoring**
  Allows continuous tracking of student progress and performance.

- **User-Friendly Interface**
  Easy to use for teachers and management without requiring technical knowledge.

- **Centralized Data Access**
  Combines data from different sources into one platform for better accessibility.

- **Error Reduction**
  Minimizes human errors that occur in manual data processing.

- **Supports Institutional Growth**
  Helps institutions improve teaching strategies and overall academic performance.

CHAPTER 5
HARDWARE AND SOFTWARE REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

The hardware requirements for developing and running the *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* are minimal, as Power BI is a lightweight and efficient tool. The following specifications are recommended:

- **Processor**
  Intel Core i3 or higher
- **RAM**
  Minimum 4 GB (8 GB recommended for better performance)
- **Storage**
  At least 500 GB hard disk or SSD
- **System Type**
  64-bit operating system
- **Display**
  Standard monitor with a resolution of 1366 × 768 or higher
- **Input Devices**
  Keyboard and mouse
- **Internet Connection**
  Required for downloading Power BI, accessing online data sources, and publishing reports

5.2 SOFTWARE REQURIMENTS

The following software components are required for developing and running the *University Student Academic Performance and Engagement Analytics Dashboard using Power BI*:

a. **Operating System**
   Windows 10 or later (64-bit recommended)

b. **Power BI Desktop**
   Used for data modeling, visualization, and dashboard creation

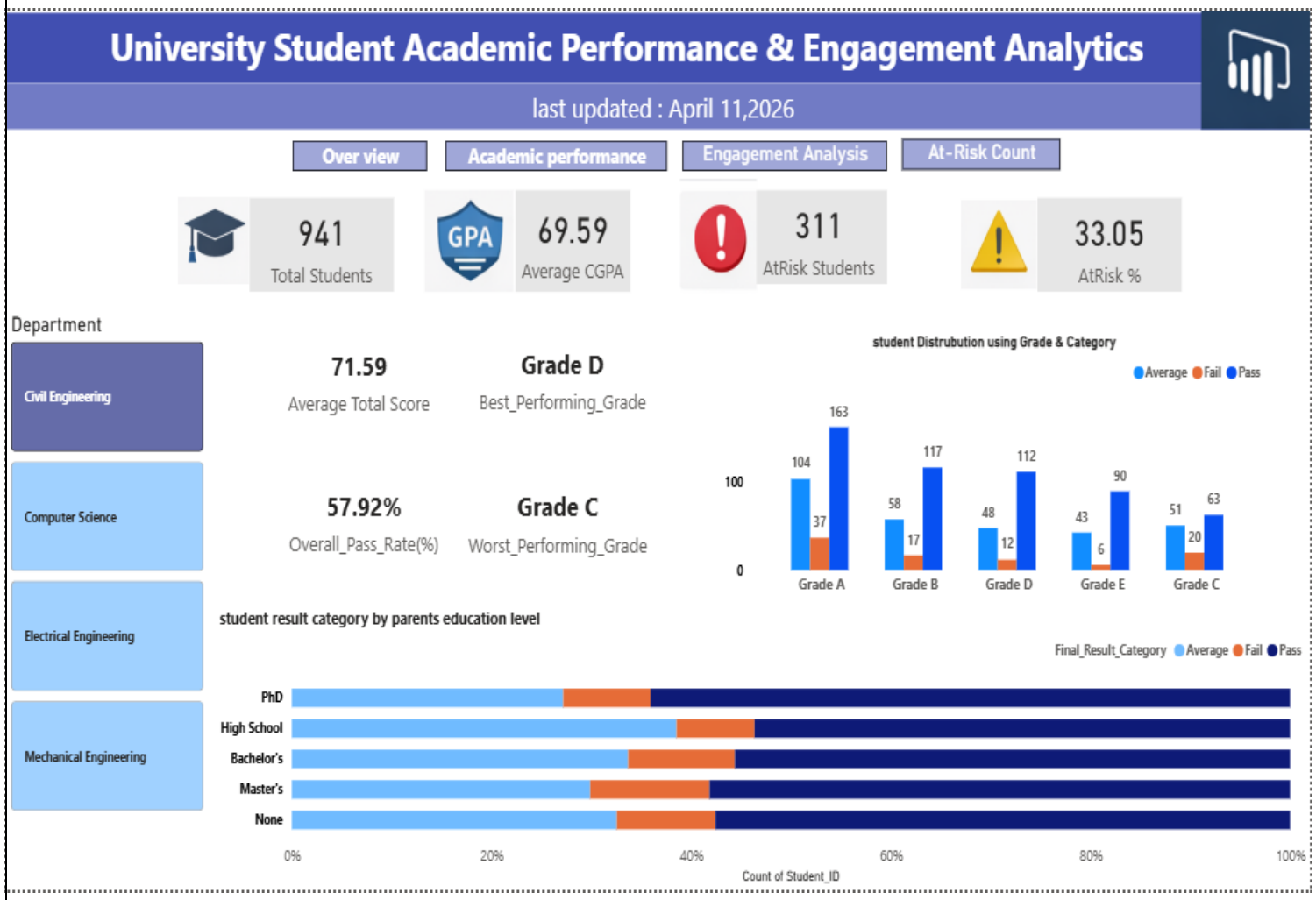
c. **Microsoft Excel**
   Used for data collection, preprocessing, and importing datasets

d. **Web Browser**
   Google Chrome / Microsoft Edge for accessing Power BI Service (online dashboards)

e. **Power BI Service (Optional)**
   For publishing, sharing, and accessing dashboards online

f. **Database (Optional)**
   MySQL / SQL Server (if data is stored in databases

CHAPTER 6
SYSTEM DESIGN


University Student Academic Performance & Engagement Analytics
last updated : April 11,2026
Over view
Academic performance
Engagement Analysis
At-Risk Count
941
Total Students
GPA
69.59
Average CGPA
311
AtRisk Students
33.05
AtRisk %
Department
Civil Engineering
Computer Science
Electrical Engineering
Mechanical Engineering
71.59
Average Total Score
Grade D
Best_Performing_Grade
57.92%
Overall_Pass_Rate(%)
Grade C
Worst_Performing_Grade
student Distrubution using Grade & Category
Average
Fail
Pass
100
0
104
37
163
58
17
117
48
12
112
43
6
90
51
20
63
Grade A
Grade B
Grade D
Grade E
Grade C
student result category by parents education level
Final_Result_Category
Average
Fail
Pass
PhD
High School
Bachelor's
Master's
None
0%
20%
40%
60%
80%
100%
Count of Student_ID

Academic Performance

Grade

Grade A | Grade B | Grade C

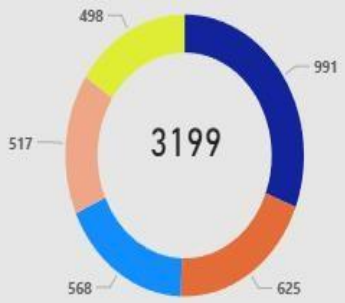
Grade D | Grade E

Student_ID	Gender	Grade	Average of Attendance (%)
Student_1	Female	Grade E	73.29
Student_10	Female	Grade E	55.44
Student_100	Male	Grade D	50.84
Student_1000	Female	Grade B	74.01
Student_1001	Female	Grade A	79.95
Student_1002	Male	Grade C	59.02
Student_1003	Male	Grade E	73.98
Student_1004	Female	Grade A	71.64
Student_1005	Male	Grade E	72.56
Student_1006	Male	Grade B	65.18
Student_1007	Female	Grade A	99.34
Student_1008	Female	Grade E	63.18
Student_1009	Female	Grade D	68.49
Student_101	Male	Grade B	62.72
Student_1010	Female	Grade E	78.74

33.32

AtRisk %

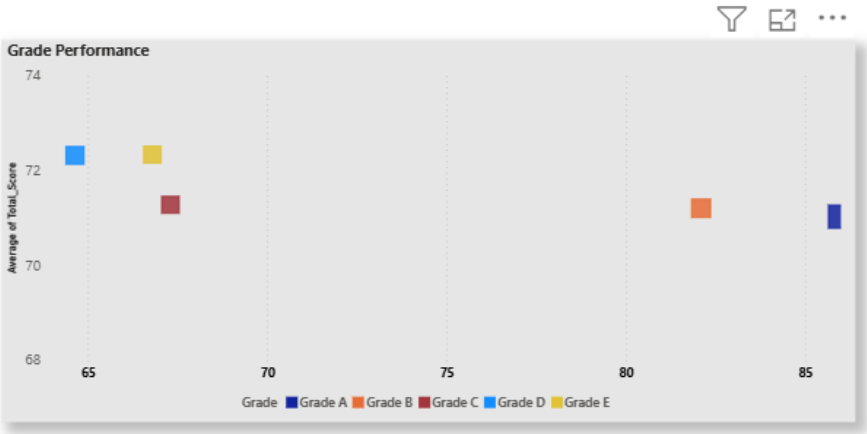
Grade wise Student count



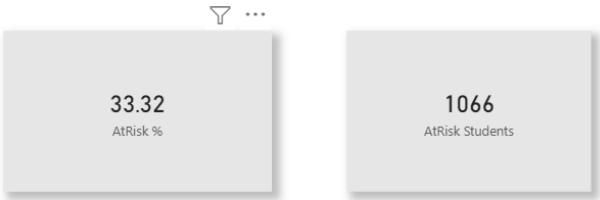
Grade ● Grade A ● Grade B ● Grade D ● Grade C ● Grade E

Aaron

Engagement Analysis



At risk analysis



i.

The system design of the *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* focuses on creating an efficient, user-friendly, and interactive platform for analyzing student data. This chapter explains how the system is organized, how data flows through it, and how different components work together to generate meaningful insights.

The design follows a structured approach that includes data collection, data preprocessing, data modeling, and data visualization.

The system design of the *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* focuses on creating an efficient, user-friendly, and interactive platform for analyzing student data. This chapter explains how the system is organized, how data flows through it, and how different components work together to generate meaningful insights.

The design follows a structured approach that includes data collection, data preprocessing, data modeling, and data visualization.

1. **Data Collection**
   The system collects data from various sources such as student academic records, attendance logs, and engagement metrics. These datasets may be stored in Excel files, databases, or other formats.

2. **Data Preprocessing**
   Raw data is cleaned and transformed to ensure accuracy and consistency. This includes handling missing values, removing duplicates, and formatting data appropriately for analysis.

3. **Data Modeling**
   In Power BI, relationships are established between different datasets to create a structured data model. Tables are linked using common fields such as student ID, course ID, etc., enabling efficient querying and analysis.

4. **Data Visualization**
   The processed data is presented using interactive dashboards. Various visual elements such as bar charts, line graphs, pie charts, and KPIs are used to display insights clearly.

5. **User Interaction**
   The system provides features like filters, slicers, and drill-down options, allowing users to explore data at different levels and customize their view.

6. **System Workflow**
   The overall workflow of the system includes:

   -Importing data into Power BI

   -Cleaning and transforming data using Power Query

   -Creating relationships and measures using DAX

   - Designing dashboards and reports

   -Publishing and sharing insights

7. Design Goals

- Simplicity and ease of use
- Clear and interactive visualization
- Efficient data handling
- Scalability for larger datasets

```
+-------------+       +-------------+             +-------------+
| Data Collection | -----> | Data Preprocessing | -----> |  Data Modeling  |
| (Excel, Database) |       | (Cleaning, Format) |           | (Relationships)  |
+-------------+       +-------------+             +-------------+
                                                          |
                                                          v
   +-------------+       +-------------+         +-------------+
| Data Visualization | <----- |  Power BI Engine | -----> |  User Interaction  |
| (Charts, KPIs)     |       | (DAX, Power Query) |          | (Filters, Slicers) |
   +-------------+       +-------------+         +-------------+
                                                          |
                                                          v
                                                   +-------------+
                                                   |  Report Sharing  |
                                                   | (Power BI Service) |
                                                   +-------------+
```

CHAPTER 7
IMPLEMENTATION

6. IMPLEMENTATION

Project Implementation of clustering using irisdataset Platform:

6.1 Implementation Overview

The implementation of the mini project "Clustering using Iris Dataset" is done in Python using libraries such as NumPy, pandas, scikit-learn, and Matplotlib/Seaborn. The system follows a typical machine learning pipeline: import libraries, load the Iris dataset, preprocess features, determine the number of clusters, apply K-Means clustering, visualize the clusters, and evaluate the results.

6.2 Importing libraries and loading dataset

- The required libraries (NumPy, pandas, Matplotlib/Seaborn, scikit-learn) are imported at the beginning of the program.
- The Iris dataset is loaded using sklearn.datasets.load_iris(), which provides the feature matrix (sepal length, sepal width, petal length, petal width) and the true species labels for evaluation.

6.3 Data preparation and preprocessing

- The feature values are stored in a pandas DataFrame for easy viewing and manipulation, and columns are named according to the Iris features.
- Optionally, the features are standardized using StandardScaler so that all four measurements have similar scale, improving the performance of distance-based clustering.
- Basic exploratory analysis (head, describe, simple plots) is done to understand the distribution of features and relationships between them.

6.4 Determining the number of clusters (Elbow Method)

- A loop runs K-Means for different values of k (e.g., 1 to 10) and stores the within-cluster sum of squares (inertia) for each k.
- An Elbow graph is plotted with k on the x-axis and inertia on the y-axis; the "elbow" in the curve helps identify the appropriate number of clusters, which for this dataset is typically k=3.

6.5 Applying K-Means clustering

- The K-Means algorithm from sklearn.cluster.KMeans is initialized with n_clusters=3 (or the chosen k), init='k-means++', and a fixed random_state for reproducible results.
- The model is fitted on the feature matrix, which computes cluster centroids and assigns each sample to the nearest centroid based on Euclidean distance

The implementation of the *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* involves the practical execution of the system design. This phase focuses on transforming raw data into meaningful insights through a series of well-defined steps using Power BI tools and techniques.

- . **Data Importing**
  Student data related to academic performance, attendance, and engagement is imported into Power BI from sources such as Microsoft Excel or databases.

- **Data Cleaning and Transformation**
  Using Power Query Editor, the data is cleaned by removing duplicates, handling missing values, correcting data types, and ensuring consistency across datasets.

- **Data Modeling**
  Relationships are created between different tables using common fields such as Student ID, Course ID, etc. This helps in organizing the data for efficient analysis .
- **Creating Measures and Calculations**
  DAX (Data Analysis Expressions) is used to create calculated columns and measures such as average marks, attendance percentage, and performance indicators.

- **Dashboard Development**
  Interactive dashboards are created using various visualizations like bar charts, line graphs, pie charts, cards, and KPIs to represent the data effectively.

- **Adding Interactivity**
  Features like slicers, filters, and drill-down options are implemented to allow users to explore data dynamically.

- **Testing and Validation**
  The dashboard is tested to ensure accuracy of data, proper functioning of visuals, and correct implementation of filters and calculations.

. Deployment and Sharing

The final dashboard is published to Power BI Service, allowing users to access and share reports online.

MESURES:

- **AtRisk %** =

```
DIVIDE(
CALCULATE(
COUNTROWS('StudentData'),
'StudentData'[Grade] IN {"Grade D", "Grade E"}
  ),
COUNTROWS('StudentData')
) * 100
```

- **AtRisk Students** =

```
CALCULATE(
COUNTROWS('StudentData'),
'StudentData'[Grade] IN {"Grade D", "Grade E"}
)
```

- **Average CGPA** = AVERAGE('StudentData'[Final_Score])

- **Average Total Score** = AVERAGE(StudentData[Total_Score])

- **Best_Performing_Grade** =

```
VAR SummaryTable =
ADDCOLUMNS(
SUMMARIZE(
StudentData,
StudentData[Grade]
),
"AvgScore",
CALCULATE(
AVERAGE(StudentData[Total_Score])
)
)

VAR TopRow =
TOPN(
1,
SummaryTable,
[AvgScore],
DESC
)

RETURN
MAXX(
TopRow,
StudentData[Grade]
)
```

- **Overall_Pass_Rate(%)** =

```
FORMAT(
DIVIDE(
CALCULATE(
COUNTROWS(StudentData),
StudentData[Total_Score] >= 70
),
COUNTROWS(StudentData)
),
"0.00%"
)
```

- **Total Students** = COUNTROWS('StudentData')

Total_Score =

```
StudentData[Midterm_Score] * 0.25 +
StudentData[Final_Score] * 0.40 +
StudentData[Assignments_Avg] * 0.15 +
StudentData[Quizzes_Avg] * 0.10 +
StudentData[Projects_Score] * 0.10
```

CHAPTER 8
RESULTS

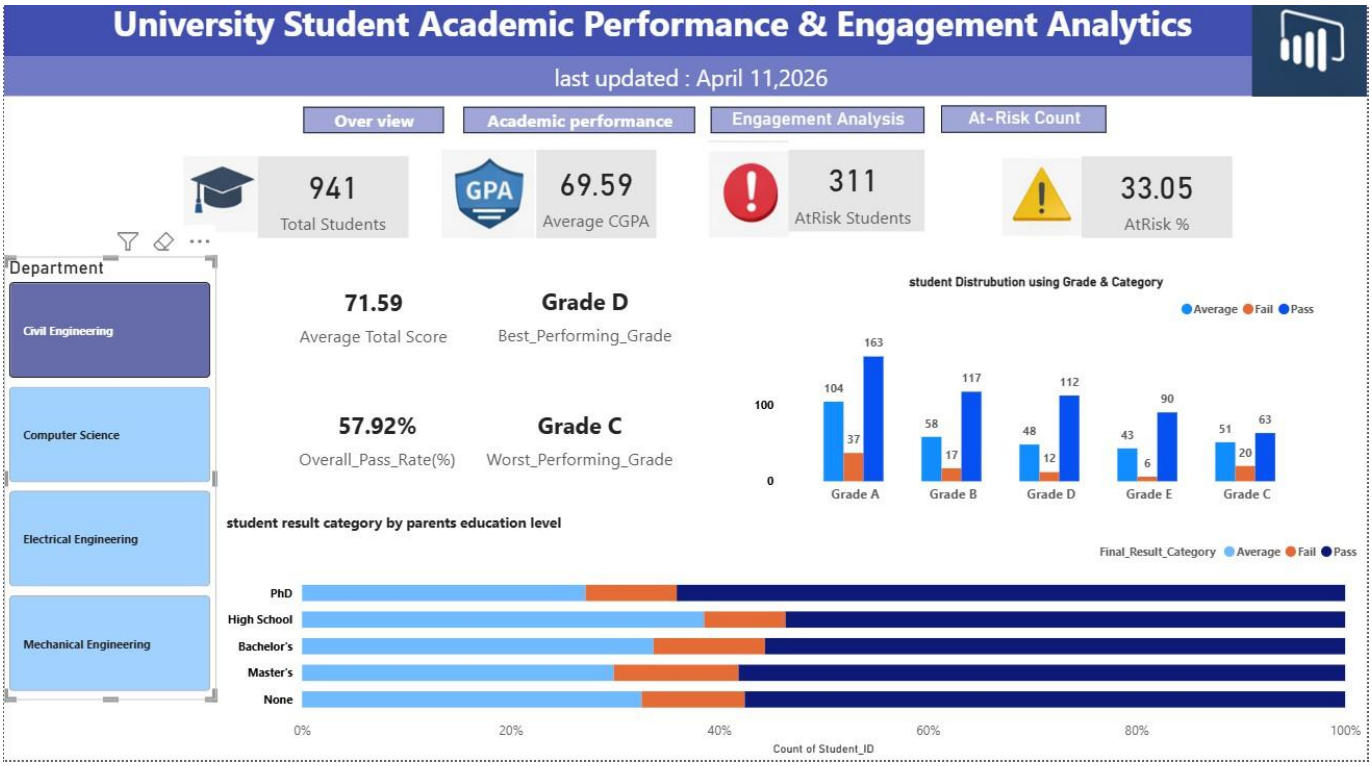
- The *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* was successfully developed and implemented. The system effectively transforms raw student data into meaningful and interactive visual insights.

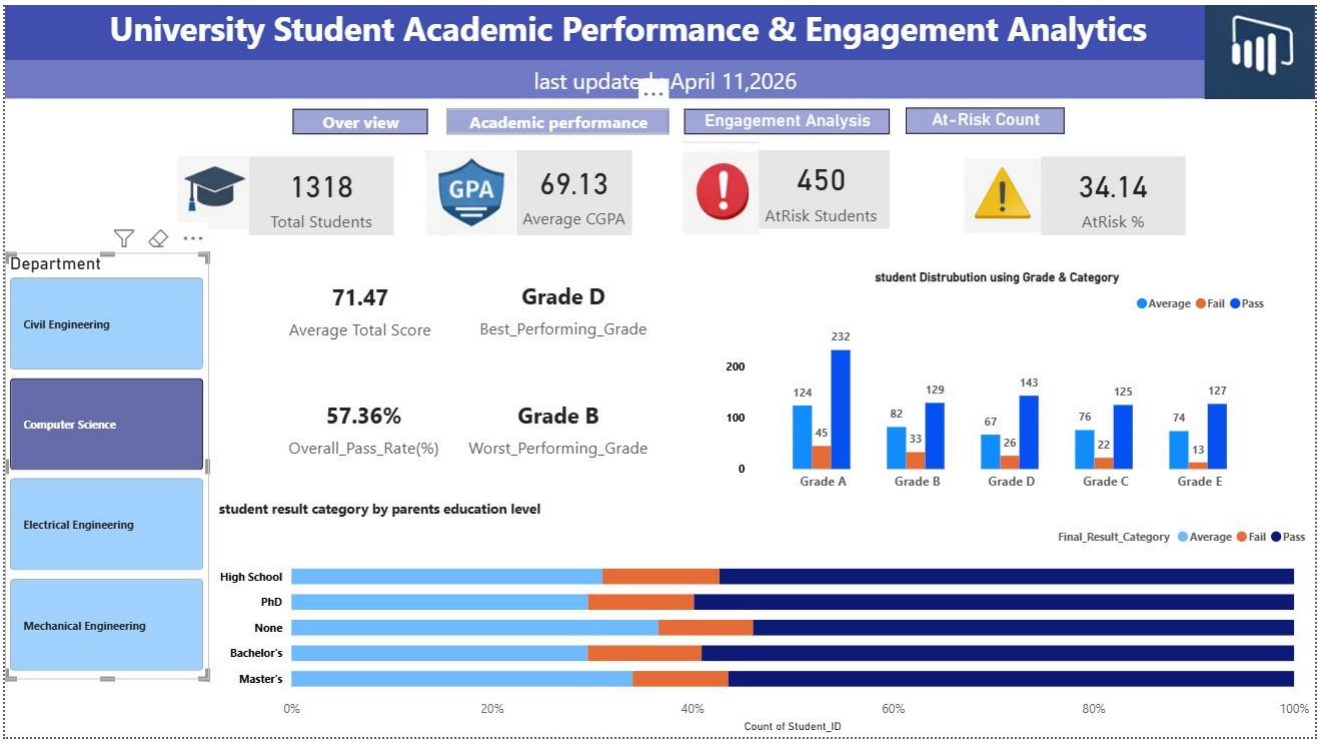
- The dashboard provides a clear overview of key performance indicators such as student marks, attendance percentage, and engagement levels. Through various visualizations like charts, graphs, and KPI cards, users can easily monitor and analyze student performance.

- The implementation of features such as filters, slicers, and drill-down capabilities allows users to explore data in detail and gain deeper insights. The system successfully identifies trends and patterns in student performance, helping educators understand factors affecting academic outcomes.

- One of the major results of the project is the ability to identify at-risk students based on low performance and attendance. This enables timely intervention and support to improve student success rates.

- The dashboard also improves efficiency by reducing manual data analysis and minimizing errors. It provides real-time or near real-time insights, allowing quick and informed decision-making.


University Student Academic Performance & Engagement Analytics
last updated : April 11,2026
Over view
Academic performance
Engagement Analysis
At-Risk Count
941
Total Students
69.59
Average CGPA
311
AtRisk Students
33.05
AtRisk %
Department
Civil Engineering
Computer Science
Electrical Engineering
Mechanical Engineering
71.59
Average Total Score
Grade D
Best_Performing_Grade
57.92%
Overall_Pass_Rate(%)
Grade C
Worst_Performing_Grade
student Distrubution using Grade & Category
Average Fail Pass
Grade A
Grade B
Grade D
Grade E
Grade C
student result category by parents education level
Final_Result_Category Average Fail Pass
PhD
High School
Bachelor's
Master's
None
Count of Student_ID


University Student Academic Performance & Engagement Analytics
last updated : April 11,2026
Over view
Academic performance
Engagement Analysis
At-Risk Count
1318
Total Students
69.13
Average CGPA
450
AtRisk Students
34.14
AtRisk %
Department
Civil Engineering
Computer Science
Electrical Engineering
Mechanical Engineering
71.47
Average Total Score
Grade D
Best_Performing_Grade
57.36%
Overall_Pass_Rate(%)
Grade B
Worst_Performing_Grade
student Distrubution using Grade & Category
Average Fail Pass
Grade A
Grade B
Grade D
Grade C
Grade E
student result category by parents education level
Final_Result_Category Average Fail Pass
High School
PhD
None
Bachelor's
Master's
Count of Student_ID

CHAPTER 9

CONCLUSION

7. CONCLUSION

- The *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* has been successfully designed and implemented to address the challenges of analyzing student data in educational institutions. The project demonstrates how raw data related to academic performance, attendance, and engagement can be effectively transformed into meaningful insights using data visualization techniques.

- By utilizing Power BI, the dashboard provides an interactive and user-friendly platform that enables educators and administrators to monitor key performance indicators, identify trends, and evaluate student progress efficiently. The integration of features such as filters, slicers, and drill-down capabilities enhances the overall usability and analytical depth of the system.

- The project also highlights the importance of data-driven decision-making in improving academic outcomes. The ability to identify at-risk students at an early stage allows institutions to take timely actions and provide necessary support, ultimately enhancing student performance and engagement.

- In conclusion, this project proves that business intelligence tools like Power BI play a vital role in modern education systems by simplifying data analysis, improving accuracy, and supporting strategic planning. The developed dashboard serves as an effective solution for academic performance analysis and can be further enhanced with advanced analytics in the future.

CHAPTER 10

FUTURE SCOPE

8. FUTURE WORK

The *University Student Academic Performance and Engagement Analytics Dashboard using Power BI* provides a strong foundation for data-driven analysis in the education sector. However, there are several opportunities to enhance and extend the system in the future.
One of the major improvements can be the integration of predictive analytics using machine learning techniques. This would help in forecasting student performance and identifying at-risk students more accurately in advance.

The system can also be enhanced by connecting it to real-time data sources such as Learning Management Systems (LMS) or online attendance systems. This would allow continuous monitoring of student activities and performance without manual data updates.
Another possible extension is the development of a web-based or mobile application that provides access to dashboards anytime and anywhere, improving accessibility for educators and administrators.

The dashboard can be further customized by incorporating role-based access control, where different users (teachers, administrators, students) can view specific data relevant to them. Additionally, advanced features like automated alerts and notifications can be implemented to inform educators about students with low performance or attendance.
Integration with other tools and platforms, such as cloud databases and big data technologies, can improve scalability and enable handling of large datasets efficiently.

Future Enhancements

- The current system provides effective analysis and visualization of student performance and engagement.

  However, it can be further improved with the following enhancements:

- **Predictive Analytics Integration**
  Implement machine learning algorithms to predict student performance and identify at-risk students in advance.

- **Real-Time Data Updates**
  Connect the dashboard to live data sources such as Learning Management Systems (LMS) for real-time monitoring.

CHAPTER 11

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9. REFERENCES

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